Which vector store should you choose?

Our demonstration uses Chroma because a persistent local collection and straightforward Python interface make the complete pipeline easy to inspect. It stores vectors alongside text and metadata, supporting similarity queries and filters.

FAISS is a strong option when you want efficient vector indexing and direct control over search structures. Its basic `IndexFlatL2` performs exact nearest-neighbour search, but you must manage original text, metadata and persistence around it. It is a vector-search library rather than a complete document application.

Qdrant becomes attractive when you want richer filtering, named vectors and integrated hybrid or multistage queries. It can run locally, but introducing another service creates configuration and maintenance work. For a personal assistant indexing a modest folder, do not migrate merely because a benchmark chart looks impressive.

Choose by your actual requirements: dataset size, filtering, incremental updates, language support, available memory and ease of backup. The best database is the one that returns the right evidence reproducibly.

Metadata turns results into evidence

Similarity alone cannot tell which document is authoritative. Store fields such as `document_id`, `title`, `page`, `updated_at`, `department`, `language` and `access_level`. At query time, filter by authorised department or current policy version before presenting passages to the model.

Do not expose restricted documents merely because the user asks a convincing question. Access decisions belong in application code before retrieval, not in a polite instruction telling the model to keep secrets. For multiple users, enforce identity-based filters server-side. We will examine hardened access controls in the deployment volume; for now, do not combine confidential departmental folders into one unrestricted demo.

Metadata also makes citations useful. "Source: policy.txt" is better than nothing; "Travel Policy, revision 4, page 6" allows a reader to verify the claim. Carry these identifiers through the entire pipeline rather than asking the model to invent them later.

Hybrid retrieval and reranking

Vector search excels at paraphrases but can miss exact identifiers: `RTX 5070`, an invoice number or an unusual clause

